WRIGHT STATE UNIVERSITY

COLLEGE OF ENGINEERING AND COMPUTER SCIENCE

Describing Data and Spotting Outliers

Descriptive statistics, distributions and charts



What is descriptive data?

Descriptive

Summarizes what the data looks like now.

A few numbers or one picture for many rows.

Examples: average, range, a histogram.

Predictive (later)

Uses the data to guess something new.

Needs a clean, understood dataset first.

Weeks 7 and 9.

Why we need descriptive statistics

See the big picture	Thousands of rows become a few numbers.
Catch bad data	An impossible maximum shows a mistake.
Compare groups	Is this group higher than that group?
Choose next steps	The shape decides which method to use.

Min, max and range

Example: hours studied by 8 students

2	4	4	4	5	5	7	9

Statistic	Formula	Our data	Tells you
Min	Smallest value	2	Lowest end
Max	Largest value	9	Highest end
Range	Max − Min	9 − 2 = 7	How spread out. Check for impossible values.

Mean (the average)

$$\text{mean} = (x_1 + x_2 + \ldots + x_n) \div n$$

2	4	4	4	5	5	7	9	(2+4+4+4+5+5+7+9) ÷ 8 = 40 ÷ 8 = 5

Tells you

A typical value, the balance point.

Watch out

One extreme value pulls the mean.

Median and mode

median = the middle value

(sorted; even count: average the middle two)

mode = the most common value

(works for labels too)

2	4	4	4	5	5	7	9

Median = (4 + 5) ÷ 2 = 4.5

Tells you: the middle. Not moved by extremes.

Mode = 4 (appears 3 times)

Tells you: the most common answer.

Trimmed mean: ignore the extremes

3	4	5	6	6	7	7	8	9	25

Sorted values. Drop the lowest and the highest (10% from each end).

Statistic	How	Value
Mean	Add all 10, divide by 10	8.0
Median	Middle of the sorted values	6.5
Trimmed mean	Drop 3 and 25, average the other 8	6.5

Std (standard deviation)

$$\text{std} = \sqrt{[\ \Sigma(x - \text{mean})^2 \div (n - 1)\ ]}$$

x	x – mean	(x – mean)²
2	−3	9
4	−1	1
4	−1	1
4	−1	1
5	0	0
5	0	0
7	2	4
9	4	16

Sum of squares = 32

Variance = 32 ÷ 7 = 4.57

std = √4.57 = 2.14

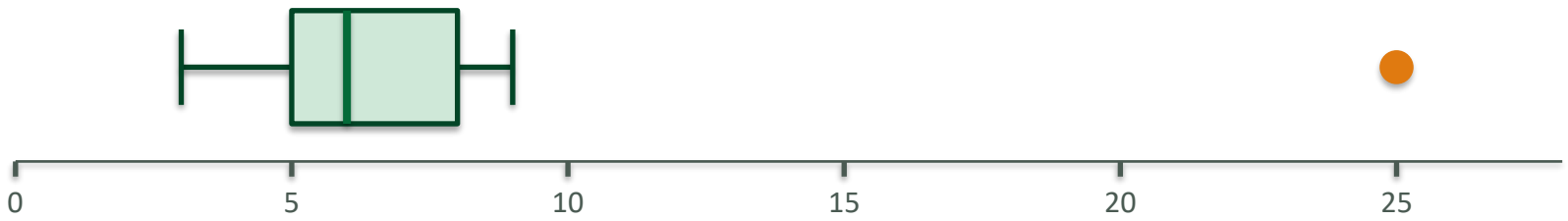
Small std: values close to the mean. Large std: spread out.

What each statistic tells you

Statistic	Gives insight into	Watch out
Min, max	Lowest and highest, impossible values	Sensitive to outliers
Range	Total spread	Uses only two values
Mean	Typical value, balance point	Pulled by extremes
Median	Middle value	Ignores the size of extremes
Mode	Most common value, also for labels	May not exist or be repeated
Std	How spread out around the mean	Also pulled by extremes

Box plot: see the outlier

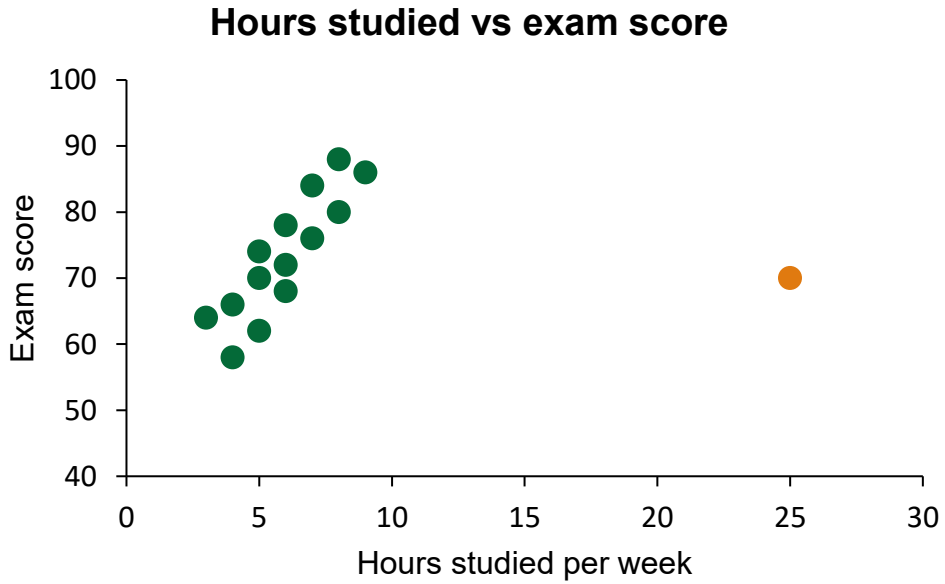
Weekly study hours of 15 students



Min	Q1	Median	Q3	Max (no outlier)	Outlier
3	5	6	8	9	25

The orange dot sits far from the box: a visual outlier.

Scatter plot: see the outlier



How to read it

Most points form a cloud that rises.

The orange point is far from the cloud.

Ask: is it a real case or a mistake?

How to calculate Q1 and Q3

3	4	4	5	5	5	6	6	6	7	7	8	8	9	25

Lower half (7 values) **Median 6** Upper half (7 values)

Q1 = 5 **Q3 = 8**

1 Sort	Put the values in order (already sorted here).
2 Split	Find the median. Split into two halves. Leave the median out.
3 Q1 and Q3	Q1 = median of the lower half. Q3 = median of the upper half.

IQR = Q3 − Q1 = 8 − 5 = 3

IQR method

IQR = Q3 − Q1

Outlier if below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$

Step	Our data
Q1 and Q3	5 and 8
IQR	8 − 5 = 3
Lower fence	5 − 1.5 × 3 = 0.5
Upper fence	8 + 1.5 × 3 = 12.5
Check 25	25 > 12.5: outlier

Not pulled by the outlier itself. Works for skewed data.

Z-score method

z = (x – mean) ÷ std

Outlier if $z > 3$ or $z < -3$ (some use 2.5)

Step	Our data
Mean and std	7.2 and 5.20
z for 25	(25 − 7.2) ÷ 5.20 = 3.42
z for 9	(9 − 7.2) ÷ 5.20 = 0.35
Check	3.42 > 3: outlier

The outlier raises the mean and std, so it can hide itself.

IQR or Z-score?

	IQR method	Z-score method
Built from	Quartiles (medians)	Mean and std
Best for	Any shape, skewed data	Bell-shaped data
Weakness	Rule of 1.5 is a convention	Outlier distorts mean and std
Companion picture	Box plot	Histogram or scatter plot

A flagged value is a question, not an answer.

What to do with an outlier

Remove

Data entry errors and glitches.

Example

25 hours typed instead of 2.5

Keep

Genuine extreme events.

Example

A real record sale or flood

Alternatives

Cap it, or transform it.

Example

Cap at the fence, or take the log

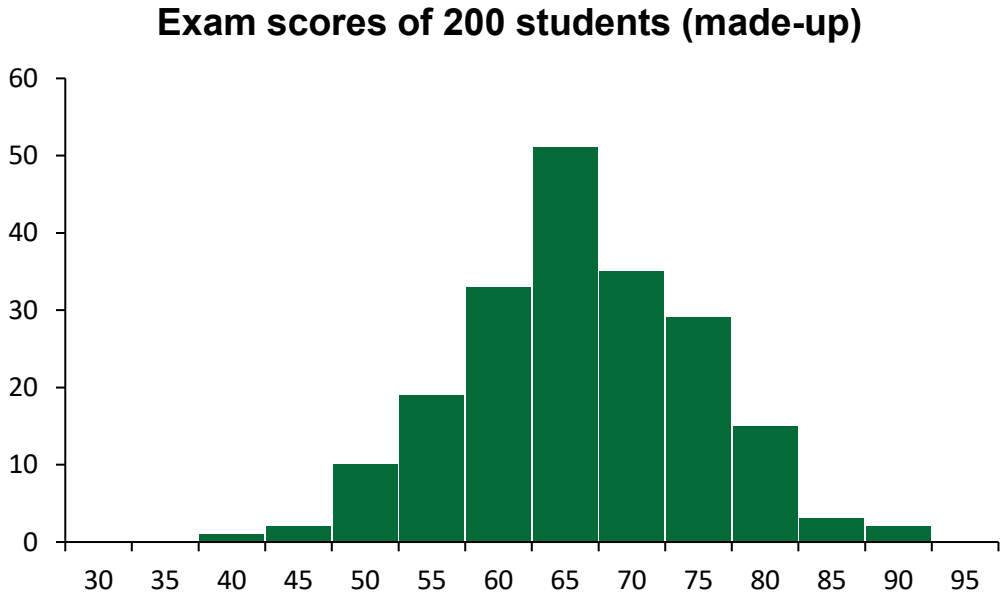
Decide by asking: could this be real? Then write it in your cleaning log.

What the outlier does to the numbers

Statistic	With the 25	Without the 25
Mean	7.2	5.9
Median	6	6
Std	5.20	1.73
Range	22	6

The median barely moves. The mean and std do.

What is a histogram?



How to read it

Each bar is a bin, like 65 to 70.

Height = how many values fall in it.

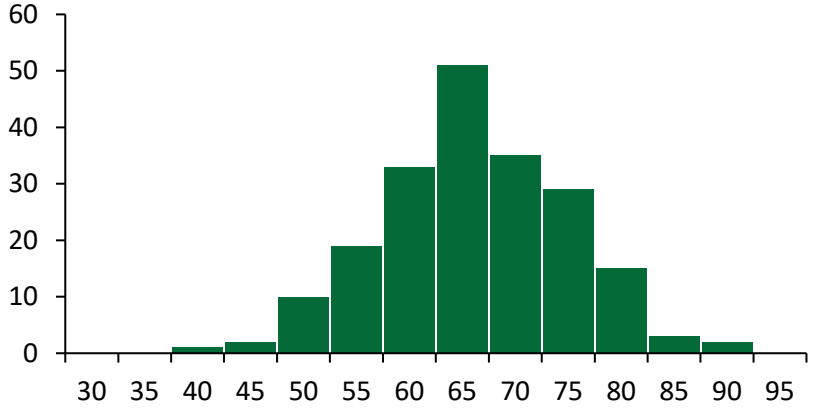
The overall outline is the distribution.

Types of distributions

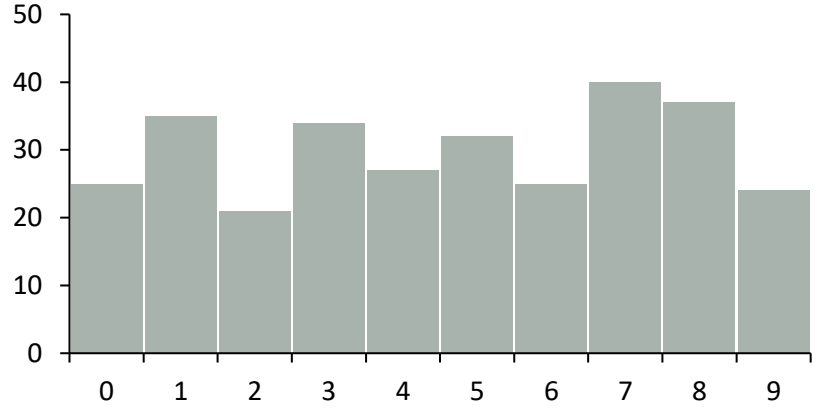
Shape	Definition	Example
Normal (bell)	Symmetric, most values near the middle	Heights, exam scores
Uniform	All values about equally likely	Rolling a die
Right-skewed	Long tail to the high side	Income, commute time
Left-skewed	Long tail to the low side	Scores on an easy exam
Bimodal	Two peaks (two groups mixed)	Heights of a mixed group
Extreme value (Gumbel)	Skewed right, models the yearly maximum	Highest flood each year

Normal and uniform distributions

Normal: bell shape



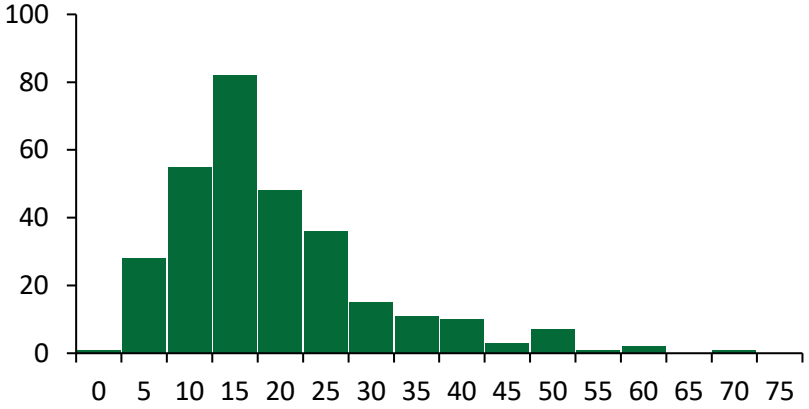
Uniform: flat



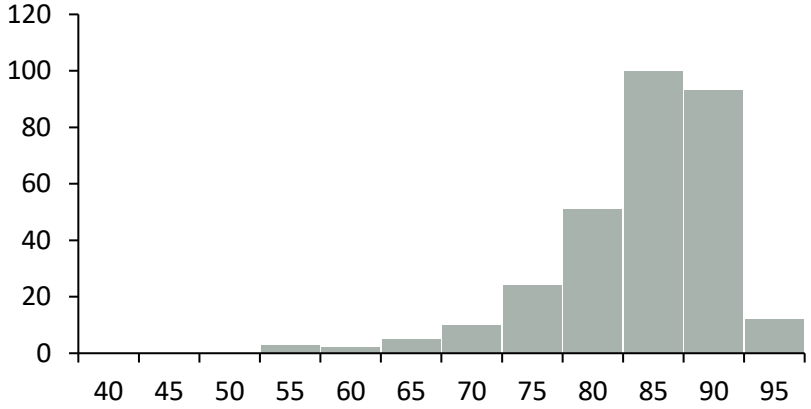
Normal: most near the middle. Uniform: all equally likely.

Skewed distributions

Right-skewed: commute minutes

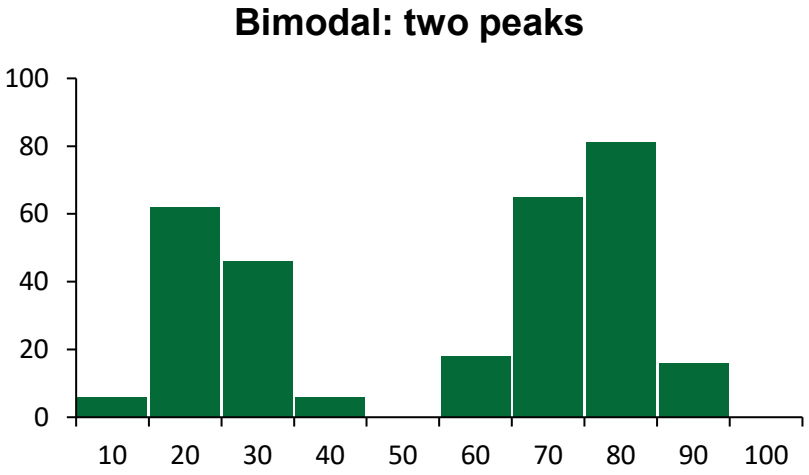


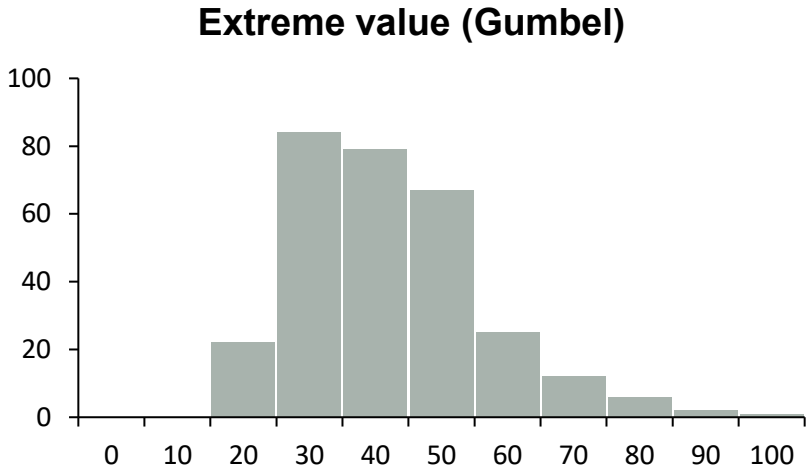
Left-skewed: easy exam scores



The tail points to the skew. Mean is pulled toward the tail.

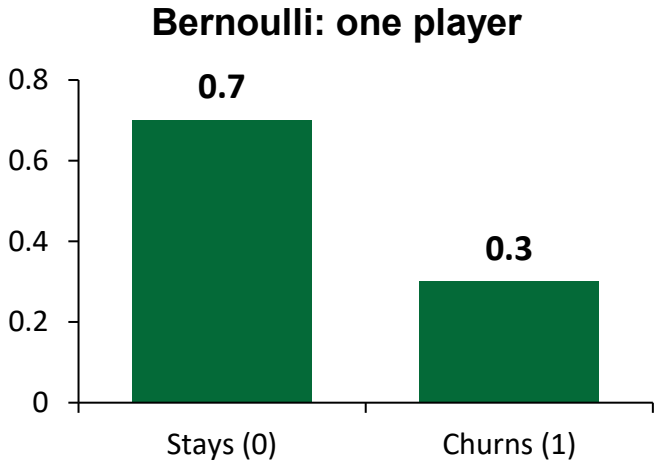
Bimodal and extreme value distributions

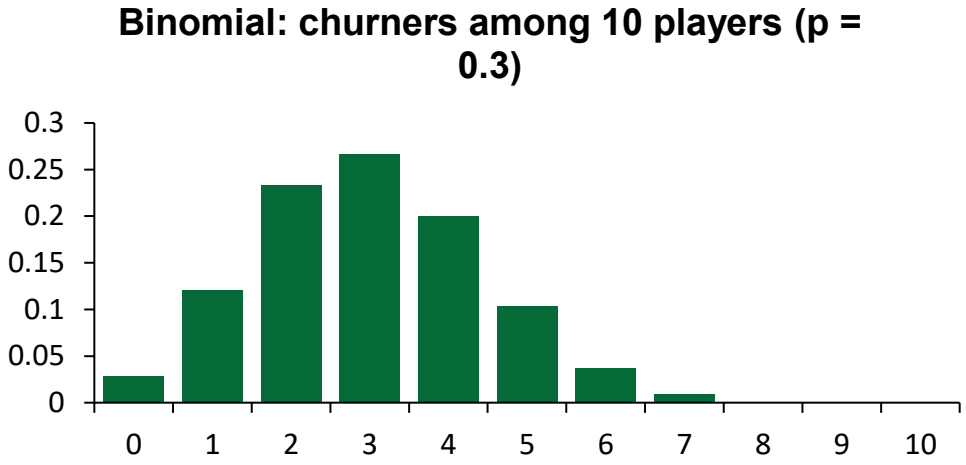




Bimodal: look for two groups. Gumbel: long right tail.

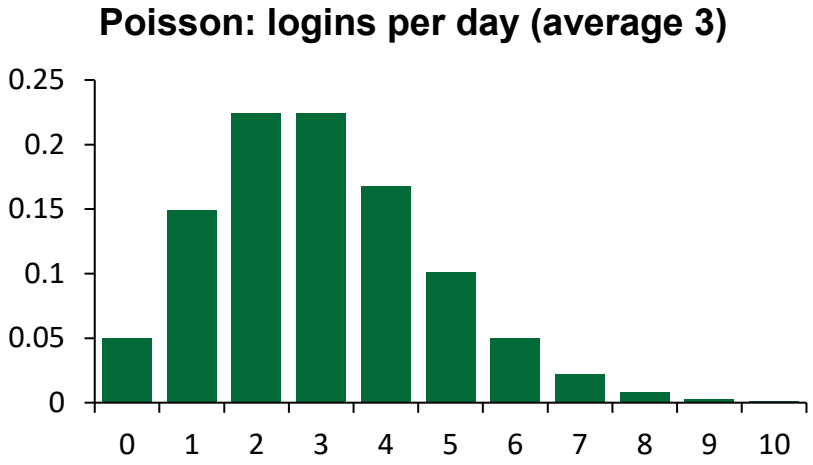
Bernoulli and binomial shapes

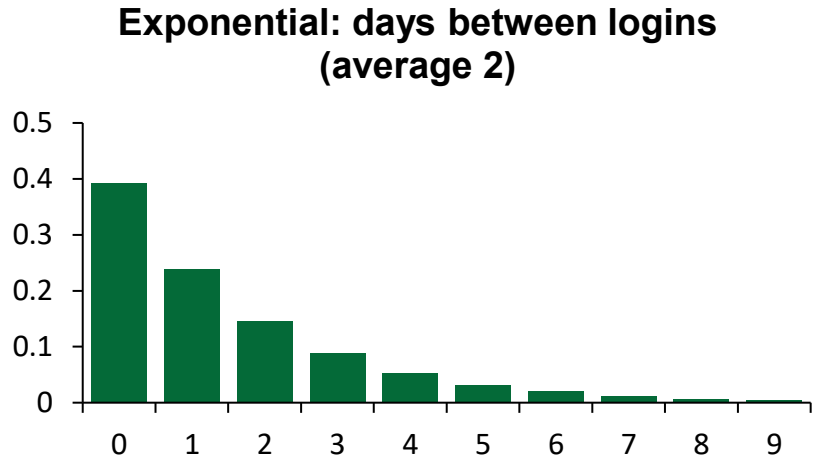




Bernoulli: one yes/no try. Binomial: how many yeses in n tries.

Poisson and exponential shapes





Poisson counts events in a window. Exponential measures the wait between them.

Right-skewed vs Gumbel

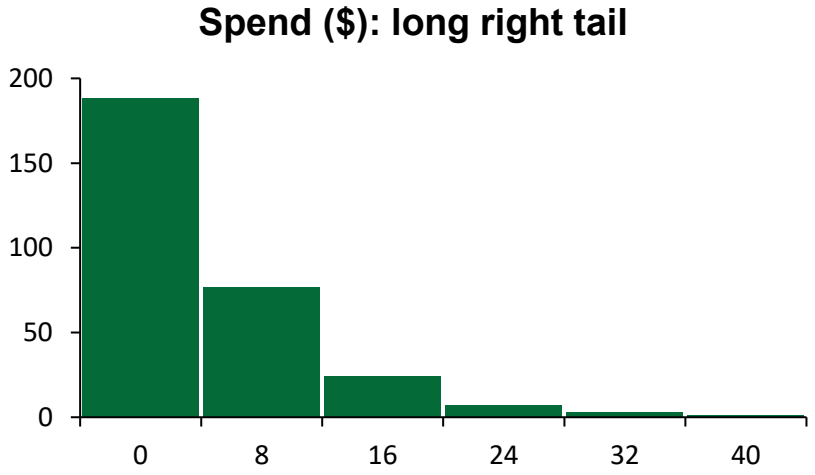
	Right-skewed	Gumbel (extreme value)
What it is	A description of a shape	A specific named distribution
Where it comes from	Many causes: income, waiting times	The largest value in each group
Example	Commute minutes	Highest river level each year
Peak	Often near the low end	Peak away from zero, then a tail
Melt Pool data	Melt pool intensity	Cooling rate

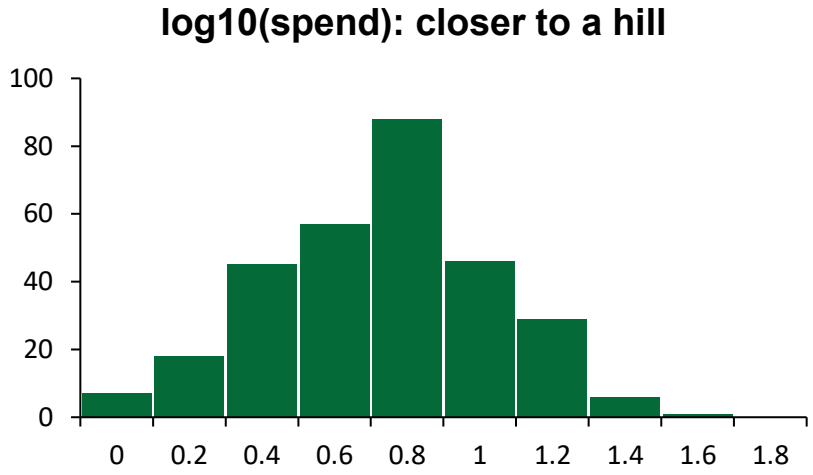
Every Gumbel is right-skewed. Not every right-skewed dataset is Gumbel.

The shape tells you which statistic to use

Shape	Mean vs median	Report
Symmetric (normal)	About equal	Mean or median
Right-skewed	Mean > median	Median
Left-skewed	Mean < median	Median
Bimodal	Middle falls between the peaks	Describe each group separately

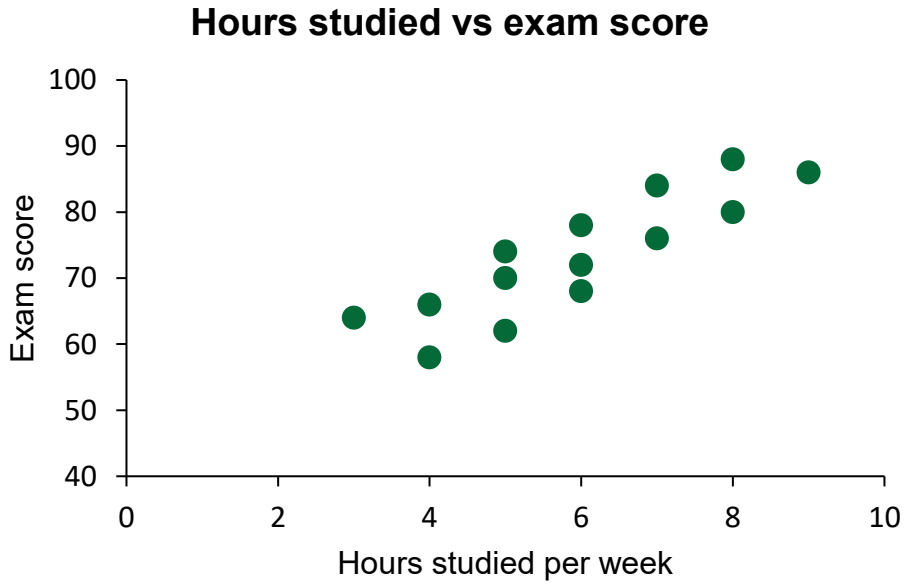
Log transform: squash a long tail





Use on positive, right-skewed columns. Keep the original.

Scatter plot and correlation



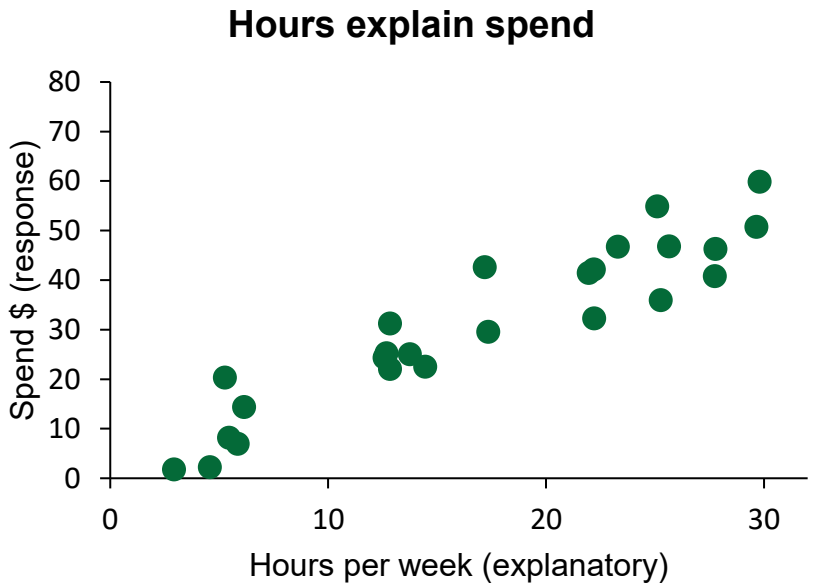
r = 0.87

Direction: positive.

Strength: strong.

−1 to +1. Near 0 means no straight-line pattern.

Explanatory vs response variable



Explanatory

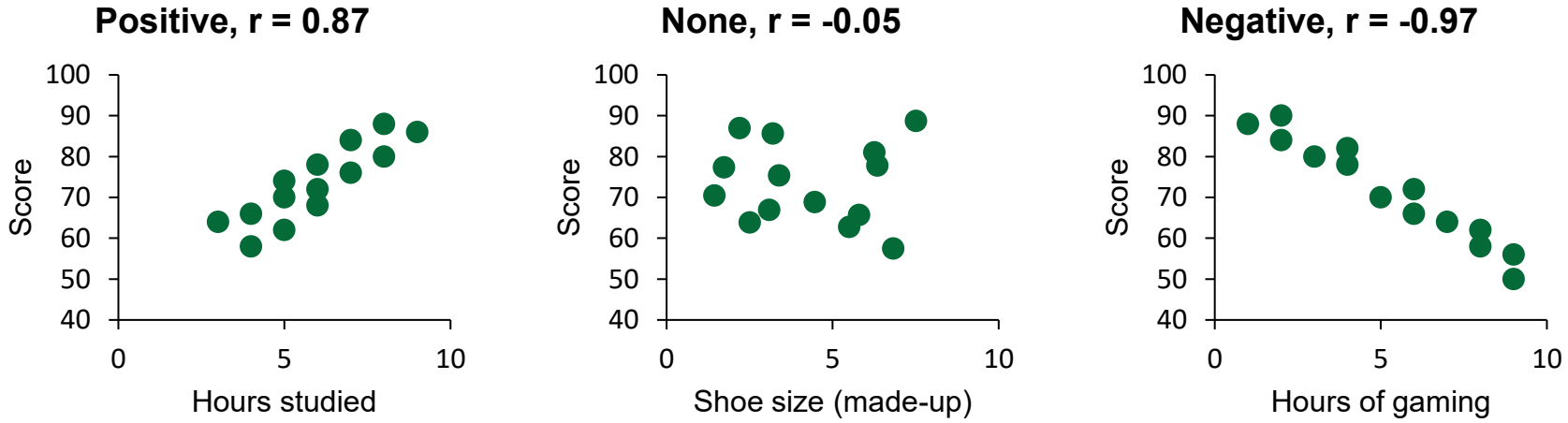
Does the explaining. Goes on the x axis.

Response

Is being explained. Goes on the y axis.

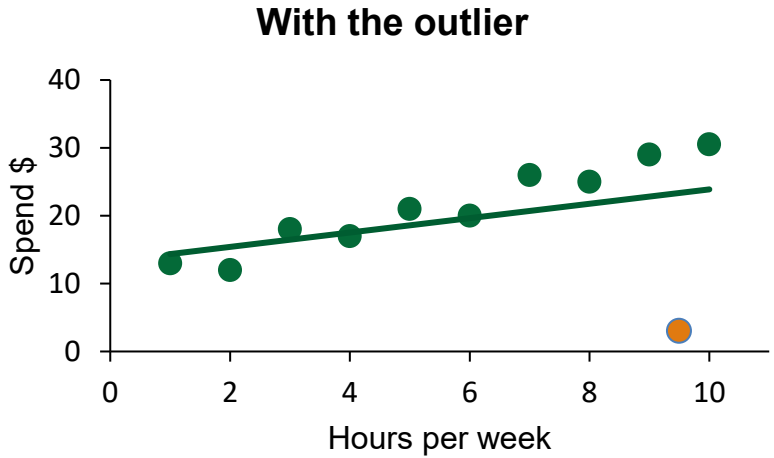
The question decides which is which.

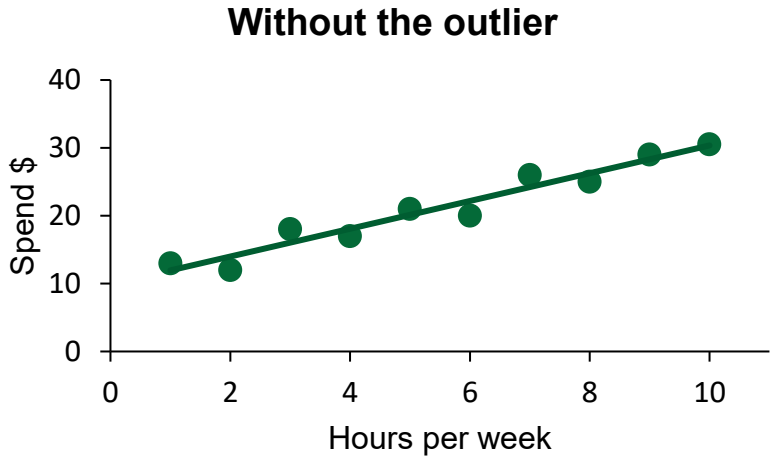
Reading the correlation number



Near +1: rise together. Near 0: no line. Near −1: one rises, the other falls.

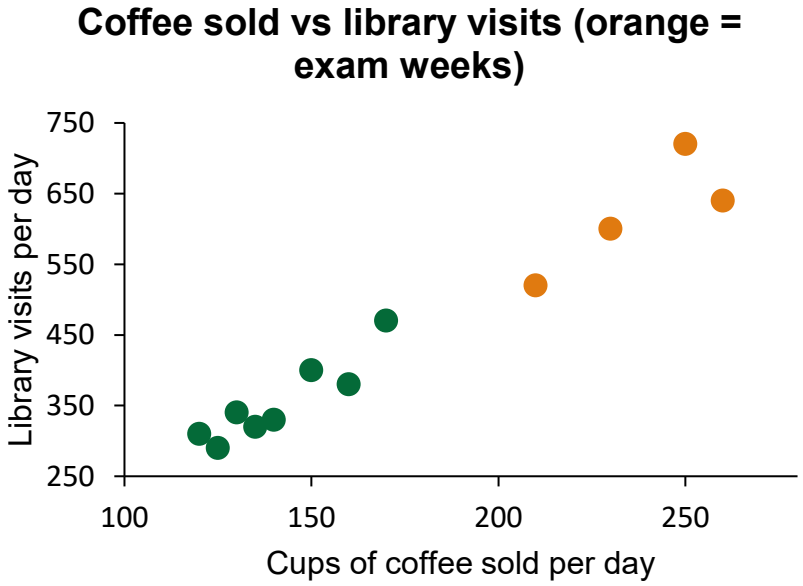
A trendline can chase an outlier





Slope with the outlier: 1.1. Without it: 2.0. Draw it both ways.

Correlation does not equal causation



r = 0.98: they rise together

Exam weeks

More coffee sold

More library visits

Coffee does not cause library visits. Exam weeks drive both.

Ask first: what is this chart for?

Compare groups	Which platform spends most?
See a shape	Is spend skewed? Any two humps?
See a relationship	Does more play mean more spend?
See change	Is churn going up over months?

Finish the sentence: "I want the reader to see ___."

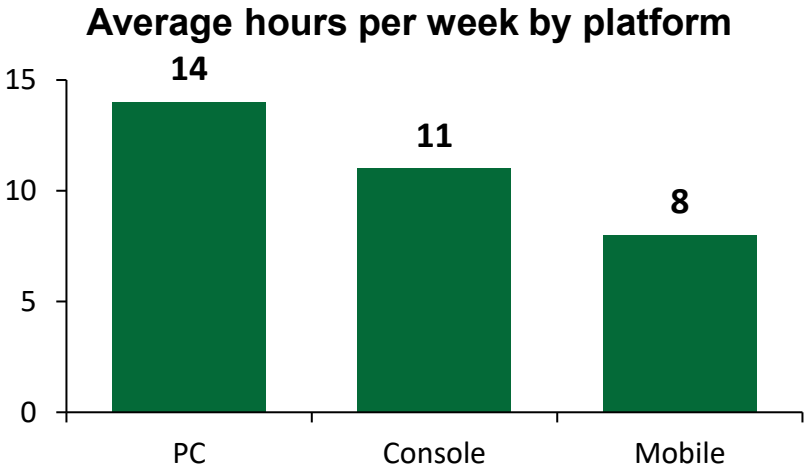
Choosing the right chart

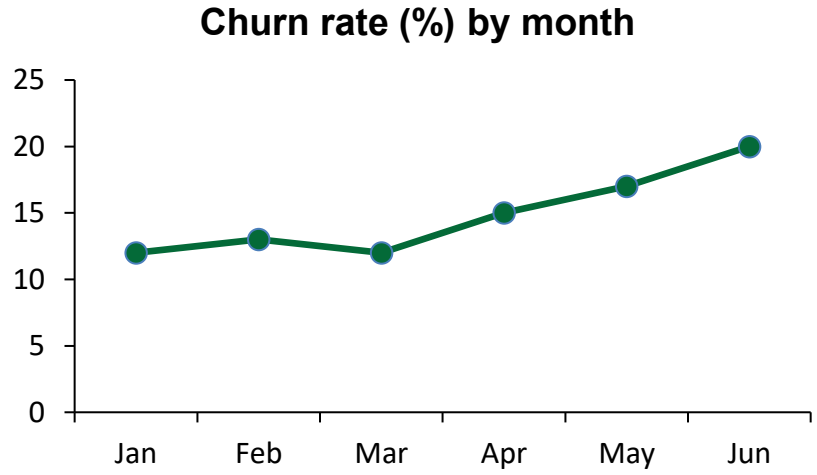
Question	Chart	Example
Compare categories	Bar chart	Sales by item
Change over time	Line chart	Temperature each day
Relationship of two numbers	Scatter plot	Hours vs score
Shape of one number	Histogram or box plot	Exam scores
Parts of a whole (few parts)	Pie or stacked bar	Share of majors

Pick the chart for the question

Question	Chart	Orange widget
What shape is one number?	Histogram	Distributions
Compare a number across groups	Box plot or violin plot	Box Plot, Violin Plot
Two numbers: related?	Scatter plot	Scatter Plot
A number over time	Line chart	Line Plot
Counts or averages by label	Bar chart	Bar Plot
Two labels together	Stacked bar or mosaic	Mosaic Display
Many numbers at once	Heat map	Heat Map

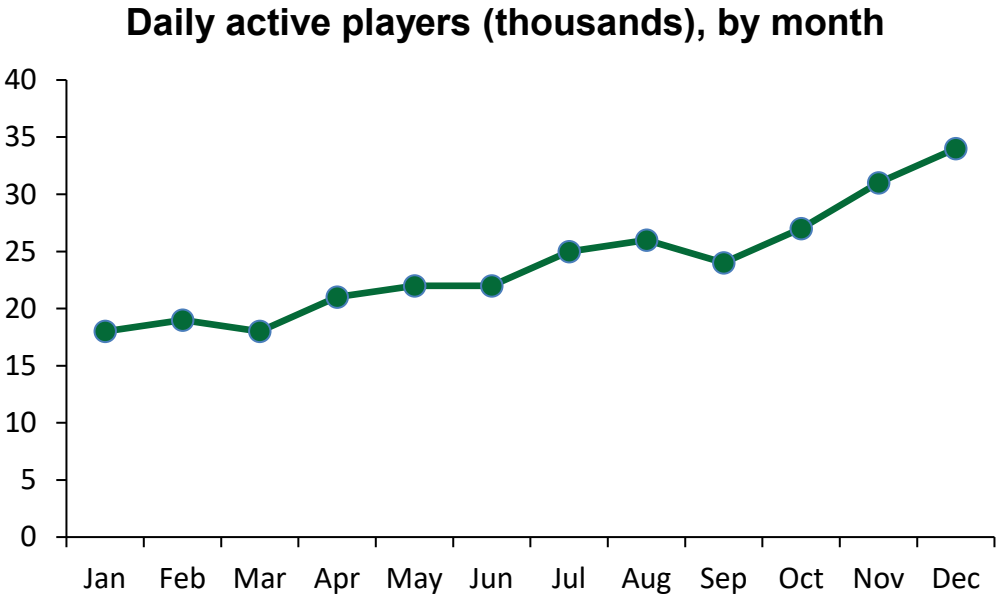
Bar chart or line chart?





Bars for separate labels. Lines for an ordered axis such as time.

Time series: data in time order



What it is

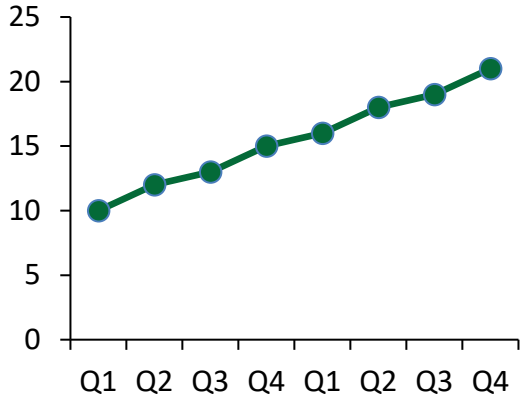
One number measured again and again, in time order.

Rules

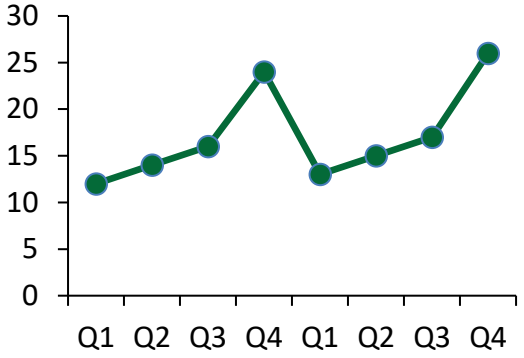
Time on the x axis, in order, with equal gaps. Use a line.

Read a time series: trend, season, spike

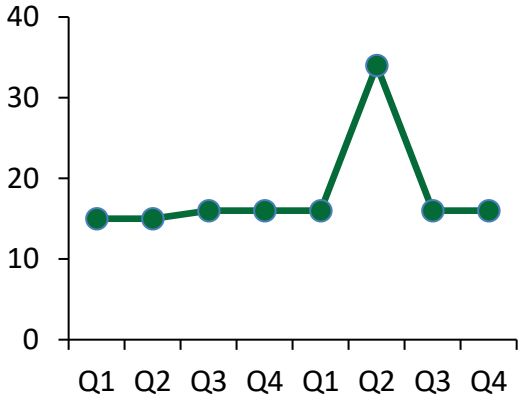
Trend: keeps rising



Season: repeats each Q4



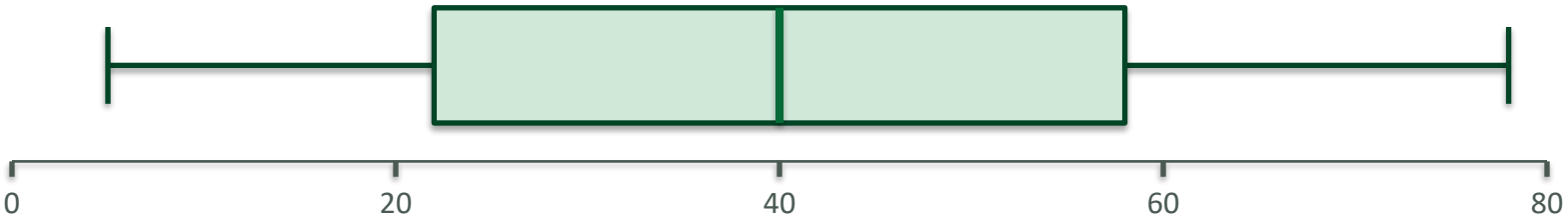
Spike: one-time jump



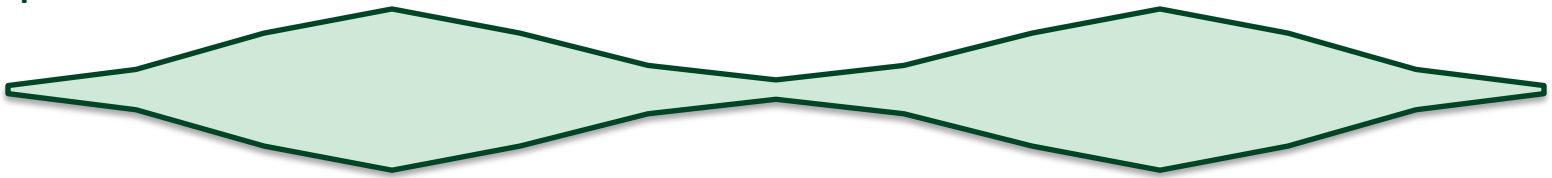
Ask: is it rising, repeating, or a one-time spike? Check a spike before you remove it.

Box plot vs violin plot

Box plot



Violin plot



Same data. The violin shows the two humps. The box hides them.

Heat map: a table with colors

	Hours	Spend	Satisfaction	Days idle
Hours	1.00	0.62	0.35	-0.48
Spend	0.62	1.00	0.28	-0.30
Satisfaction	0.35	0.28	1.00	-0.52
Days idle	-0.48	-0.30	-0.52	1.00

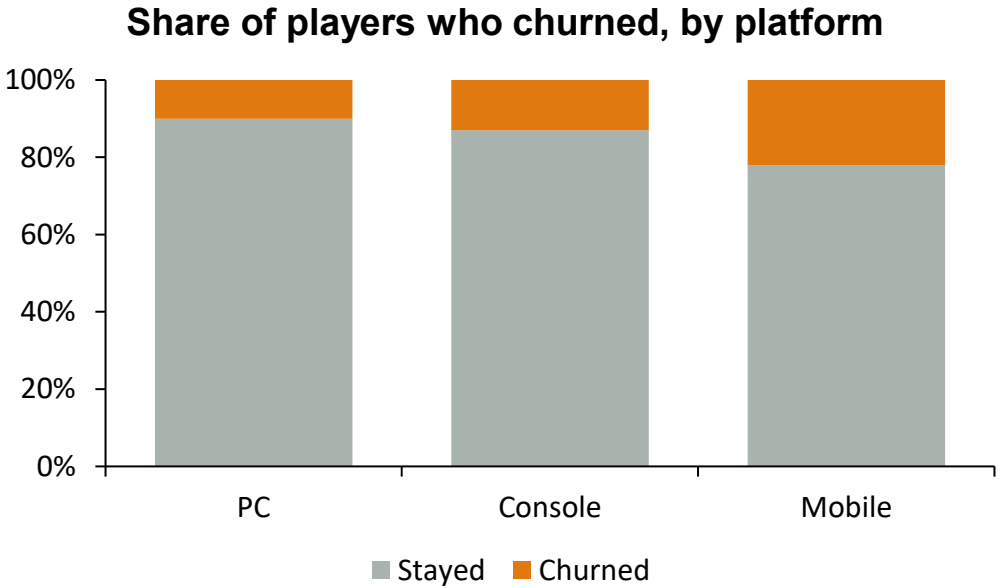
How to read it

Dark green: move together.

Dark amber: move opposite.

Pale: little relationship.

Two labels: stacked bars



Use it when

Both columns are labels.

You ask: does one label depend on the other?

Here: churn is higher on Mobile.

Traditional vs dynamic graphs

Compare	Traditional (static)	Dynamic (interactive)
What it is	A fixed picture	A graph you can act on
Viewer can	Look	Hover, zoom, filter, select
Best for	Reports, print, one message	Exploring, checking, asking new questions
In Orange	Save the plot as an image	Select points in Scatter Plot and see them in a Data Table

Same data. The difference is what the viewer can do.

When a dynamic plot is better

Situation	Better choice	Why
Hundreds of points, outliers to identify	Dynamic	Hover or select shows which rows they are
Comparing many groups	Dynamic	Filter one group at a time
Exploring data you do not know yet	Dynamic	New questions need new views
One clear message in a report	Static	The reader should not have to work
Printed page or slide	Static	Nothing to click

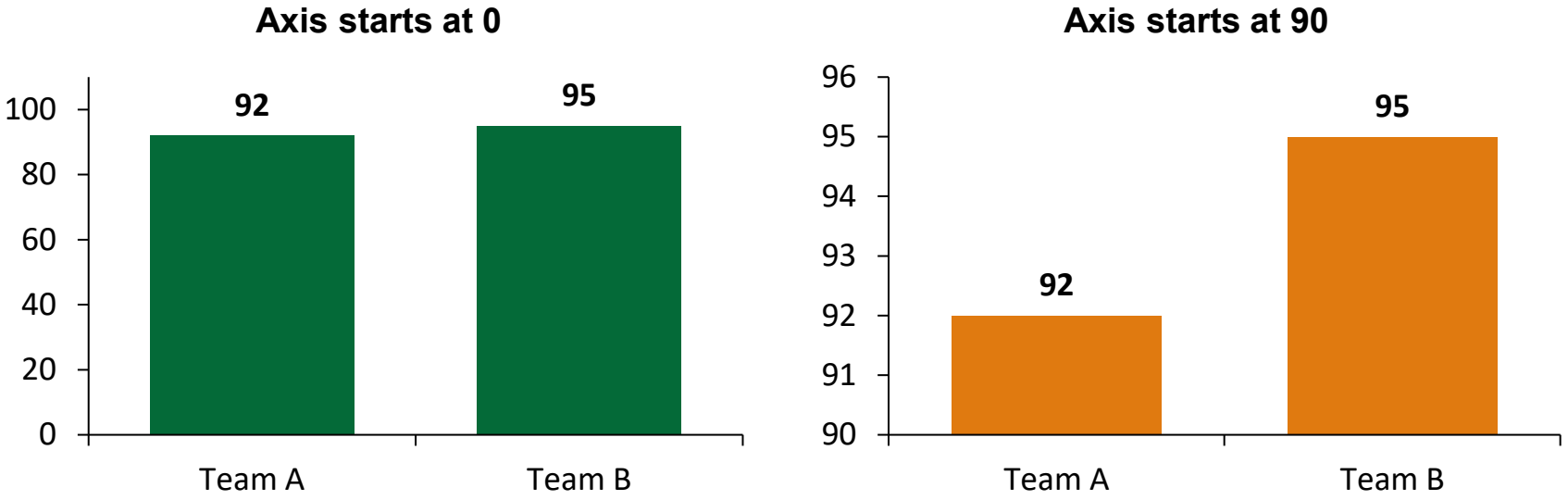
Explore dynamically. Publish one static chart.

Interactive widgets in Orange

Widget	What you can do
Scatter Plot	Hover for a row, zoom, select points and send them on
Box Plot	Hover to read Q1, median and Q3; split by a label
Violin Plot	Hover over the shape; split by a label
Heat Map	Select a block of rows or columns and send it on
Data Table	Shows the rows you selected in another plot

Select in a plot. See the rows in a Data Table.

Bad graph 1: the truncated axis



Same data. Right looks like a big gap: B is more than double A.

Bad graph 2: other tricks

3D tricks

Perspective makes front items look bigger.

Example: in a 3D pie, a 25% slice in front looks bigger than a 30% slice at the back.

Misleading scales

Different or uneven axes for the same data.

Example: sales of 900 to 1,000 look like a jump on one axis and flat on another.

Cherry-picked range

Show only the part that fits the story.

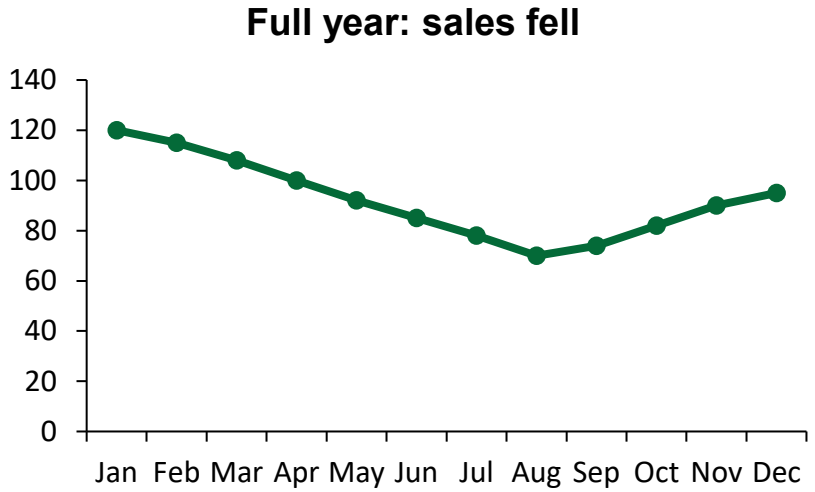
Example: show the best 4 months and hide 8 months of decline (next slide).

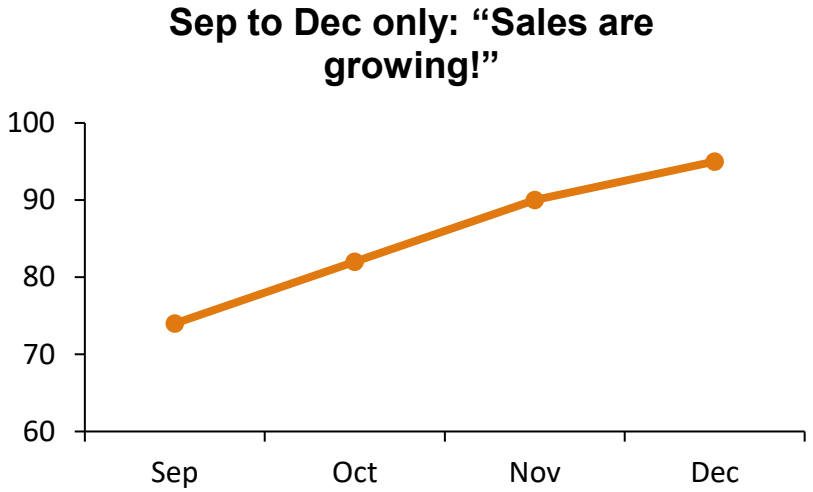
Clutter

Too much on one chart.

Example: 10 colors, no title, no units. Nobody knows where to look.

Bad graph 3: cherry-picked range





Both charts use true numbers. Ask: what period is shown, and why?

Before you share a chart

Right chart?	Does the chart type match the question?
Honest axis?	Do bars start at zero? Are ticks even?
Labeled?	Title, axis names and units.
Simple?	Few colors, no 3D, no clutter.
Sourced?	Say where the data came from.